



**Comments Submitted in Response to:
National Science Foundation
Request for Information on the Development of an
Artificial Intelligence (AI) Action Plan**

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Executive Summary

An effective national AI policy should ensure uniform value across all infrastructure components essential for American AI leadership. High-speed, low-latency fiber optic connectivity is crucial for transmitting vast amounts of data required for AI training and real-time processing. Reliable network infrastructure ensures seamless communication between data centers, edge devices, and cloud platforms, enabling efficient AI operations and innovation.

Zayo has a long history of building high-performing fiber optic networks and managing a massive global fiber optic network infrastructure. Founded in 2007, we have assembled a large portfolio of fiber networks with more than 18M miles of fiber and 146,000 route miles globally. As the only national fiber carrier to be awarded Enabling Middle Mile Broadband Infrastructure Program (MMGP) grant funds under the National Telecommunications and Information Administration (NTIA), Zayo is deploying network connectivity in rural areas while also connecting major metropolitan areas in anticipation of AI demand. Zayo was the first to break ground on an NTIA-funded Middle Mile project in June of 2024.

In this response to the National Science Foundation RFI for Artificial Intelligence Action Plan in support of the Presidential Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence), Zayo will highlight the importance of including fiber infrastructure deployments in the administration's policy considerations in securing American leadership in Artificial Intelligence (AI).

High Priority Policy Actions to Include in the AI Plan

There are opportunities to implement policies that encourage the rapid development of fiber networks in response to increased AI demand. Traditionally, middle-mile networks directly serve large bandwidth consumers and provide a bridge for last-mile connections to rural residents and businesses. AI workloads require the movement of massive datasets between data centers, edge locations, and cloud platforms. These middle-mile networks act as AI corridors to ensure ultra-high-speed, low-latency connectivity to support these data-intensive operations efficiently. Additionally, middle-mile fiber expansion inherently connects rural areas which today are experiencing high data-center growth due to power capacity. From this perspective, the middle-mile fiber provides both the underlying infrastructure to support sustainable rural economic growth through AI data center development and increases the number of rural households connected to the internet. The reallocation of existing programs, like BEAD, can prioritize funding AI corridors in rural areas.

Federal investment into middle-mile networks, or AI corridors, can lower the cost for last-mile rural connectivity and create a reliable infrastructure backbone for the critical development of AI. Additionally, an open-access policy implemented by Internet Service Providers like Zayo should be promoted at the federal level. The Institute for Local Self-Reliance defines “open-access” as “an arrangement in which one network is open to independent Service Providers to offer services”. As AI applications expand, open-access fiber networks allow multiple providers to compete and innovate, fostering infrastructure growth and ensuring that AI data centers and edge computing facilities have the connectivity and resiliency needed to scale efficiently.

The evolution of data center development, driven by power availability rather than traditional population centers, highlights the need for federal policies that support strategic infrastructure investments. Federal energy policies should align with AI infrastructure needs by prioritizing power availability for AI data centers. As power availability becomes the primary driver of data center placement, the government should streamline permitting processes for energy projects that directly support AI infrastructure. Establishing federal incentives for the expansion of energy grids in power-rich but underdeveloped regions will encourage sustainable growth.

Zayo is already adapting to these changes by developing critical middle-mile, or AI corridor, infrastructure, such as the Reno-to-Las Vegas route, to support AI data center hubs. Expanding such initiatives through federal support would enhance connectivity between key AI regions and strengthen America's AI competitiveness. Support for tax incentives or grants aimed at infrastructure development in non-traditional AI regions, such as Nevada and West Texas, would encourage investment and reduce the strain on existing metropolitan power grids.

Removal of Burdensome Requirements

When constructing fiber networks, Zayo Group and other fiber infrastructure companies must navigate various permitting requirements at the local, state, and federal levels. The construction of fiber networks, and specifically the burying or trenching of fiber cable, is the most costly element of fiber infrastructure deployment. Permits are essential for ensuring compliance with public safety, environmental regulations, and right-of-way (ROW) guidelines while minimizing disruptions to existing infrastructure and utilities. New policy can encourage collaboration between utilities and fiber infrastructure networks to optimize construction, right-of-way, and permitting costs.

There are also opportunities to enhance coordination and efficiency, particularly for fiber networks spanning multiple states that support AI infrastructure. Currently, permitting agencies operate independently, requiring sequential approvals that extend project timelines. Greater federal coordination across state lines could streamline approvals, ensuring critical fiber infrastructure is deployed efficiently. Improved agency responsiveness could prevent delays, as routine permit inquiries can take months. More direct tribal engagement could also facilitate smoother communication while respecting cultural considerations.

The federal government is well-positioned to facilitate infrastructure development on federal lands and across state jurisdictions. By fostering a more coordinated, federally supported approach—especially for fiber networks essential to AI development—the U.S. can accelerate deployment. Streamlining environmental, cultural, and archaeological reviews can leverage new technologies, such as drones, to reduce manual, time-consuming processes. Expedited ROW access can be achieved by implementing shot clocks for approvals, limiting permitting corrections, using actual-cost based permitting, minimizing reviews in previously disturbed areas, and consolidating multi-level approvals to cut costs.

At the state level, clarifying lead agency roles and refining policies could help reduce bottlenecks. Reliance on third-party contractors has increased costs, while Department of Transportation (DOT) processes—such as staff shortages and complex design requirements—have added delays. Addressing these challenges would improve deployment efficiency.

Zayo's Actions in Anticipation of AI Demand

Middle-mile networks connect regional areas to broader internet infrastructures, enabling the efficient transfer of large data volumes. Distributed systems of data centers that train, generate, and apply AI applications across the nation require AI corridor networks to provide high-capacity connections over long distances, linking cities and regions. They are essential for supporting the expansion of AI-driven data centers and meeting growing bandwidth demands.

Zayo is building 5,000+ miles of long-haul and AI corridors to support AI-driven data center growth, which is expected to increase up to six fold by 2030. However, the current rate of fiber infrastructure expansion is not keeping pace with the rapid rise in AI workloads, which could create a significant infrastructure gap, hindering the innovation and scalability of AI applications. As one of the few companies to deploy new long-haul networks in the past decade, Zayo's expansion is critical to preventing this gap.

New fiber routes are essential to support AI-driven demand. Without expansion, network constraints will limit the full potential of AI adoption and existing federal funding plays a key role in expansion. Through the NTIA's Enabling Middle Mile Broadband Infrastructure Program (MMGP), Zayo was awarded \$92.9 million to build critical middle-mile connections across eight states. These routes will lower costs for carriers to extend last-mile services, improving connectivity for rural communities, businesses, and community anchor institutions. Additionally, Zayo has designed these middle-mile routes as AI Corridors, recognizing the shift of data centers to rural areas where power capacity and land remain available.

Zayo predicts that U.S. data center power demand is expected to grow from 25 GW to between 65 GW and 148 GW by 2030, driven by AI. To meet growing demand, Zayo will add five new long-haul routes and overbuild seven existing ones, optimizing connectivity between key data center hubs. AI providers are shifting to emerging markets to access lower power costs, making alignment between fiber and energy availability critical. Several of these projects receive NTIA MMGP funding, with additional investments planned. In the past five years, Zayo has completed 15 long-haul routes totaling 5,000+ miles. Federal coordination and investment in fiber infrastructure are essential to ensuring the U.S. can meet AI's growing bandwidth needs while expanding broadband access to rural communities.

The map below depicts Zayo's North American fiber network that connects major metropolitan areas and rural communities where AI-driven data center growth is expected to increase.

